

Mohamed Abdelhack

POSTDOCTORAL FELLOW · KREMBIL CENTRE FOR NEUROINFORMATICS, CAMH

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Research Summary

I work at the intersection of computational and cognitive neuroscience, artificial intelligence, and psychiatry to unravel the neural underpinnings of perception, cognition, and mental health and how they relate to physical health. My research leverages deep learning and advanced data analytics to model brain function. I build models that reveal neural relationships that cover different transdiagnostic symptoms of mental illness. With an interdisciplinary background and a commitment to developing integrative, biologically plausible models, I aim to bridge computational theory with clinical application and advance our understanding of brain function in both health and disease.

Education

Kyoto University

PHD INFORMATICS

Kyoto, Japan

2015 - 2019

Okinawa Institute of Science and Technology

MSC SCIENCE AND TECHNOLOGY

Okinawa, Japan

2012 - 2015

Alexandria University

BSC ELECTRICAL ENGINEERING

Alexandria, Egypt

2008 - 2012

Professional Experience

- 2021- **Postdoctoral Research Fellow**, Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health
- 2019-2021 **Postdoctoral Research Assistant**, Department of Anesthesiology, Washington University in St. Louis
- 2018-2019 **Researcher**, Graduate School of Medicine, Kyoto University
- 2015-2018 **Research Assistant**, Graduate School of Informatics, Kyoto University
- 2015-2018 **Student Intern**, Department of Neuroinformatics, Advanced Telecommunication Research Institute
- 2011 **Summer Intern**, Laboratory of Neural Injury and Repair, Wadsworth Center

Publications

* equal contribution; + mentored student

SELECTED PUBLICATIONS

- Abdelhack, Mohamed**; Zhukovsky, Peter; Milic, Milos; Harita, Shreyas; Wainberg, Michael; Tripathy, Shreejoy; Griffiths, John; Hill, Sean and Felsky, Daniel. 2023. Opposing brain signatures of sleep in task-based and resting-state conditions. *Nature Communications*. URL: <https://www.nature.com/articles/s41467-023-43737-7>
- Abdelhack, Mohamed** and Kamitani, Yukiyasu. 2018. Sharpening of hierarchical visual feature representations of blurred images. *eNeuro*. URL: <https://www.eneuro.org/content/5/3/ENEURO.0443-17.2018.short>
- Abdelhack, Mohamed**; Zhang, Jiaming; Tripathi, Sandhya; Fritz, Bradley A; Felsky, Daniel; Avidan, Michael S; Chen, Yixin; and King, Christopher R. 2023. A modulation layer to increase neural network robustness against data quality issues. *Transactions on Machine Learning Research*. URL: <https://openreview.net/forum?id=MRLHN4MSmA>

PUBLISHED

- Felsky, Daniel and **Abdelhack, Mohamed**. 2025. Connecting Mechanistic Evidence for Depression, Anxiety, and Insomnia with Neuroimaging. Invited Editorial: *Sleep Medicine Reviews*
- King, Christopher R et al. (**Abdelhack, Mohamed** as a part of the TECTONICS Study Group). 2025. Effect of Telemedicine Support of Intraoperative Anesthesia Care on Postoperative Outcomes: The TECTONICS Randomized Clinical Trial. *British Journal of Anaesthesia*
- Fritz, Bradley A; King, Christopher R; **Abdelhack, Mohamed**; et al. 2024. Effect of Machine Learning on Anaesthesiology Clinician Prediction of Postoperative Complications: The Perioperative ORACLE Randomised Clinical Trial. *British Journal of Anaesthesia*
- Abdelhack, Mohamed***; Tripathi, Sandhya*; Chen, Yixin; Avidan, Michael S; and King, Christopher R. 2024. Social vulnerability and surgery outcomes: A cross-sectional analysis. *BMC Public Health*
- Akbarian, Niki; Ebrahimi, Mahbod; Dos Santos, Fernanda C; Sadat Afjeh, Sara; **Abdelhack, Mohamed**; Sanches, Marcos; Diaconescu, Andreea O; Rajji, Tarek K; Felsky, Daniel; Zai, Clement and Kennedy, James L. 2024. Examining the Role of Neuroticism Polygenic Risk in Late Life Cognitive Change: A UK Biobank Study. *Behavioral Sciences*
- Dolling-Boreham, Roberta M⁺; Mohan, Akshay; **Abdelhack, Mohamed**; Elton-Marshall, Tara; Hamilton, Hayley A; Boak, A; Felsky, Daniel. 2024. Identifying Psychosocial and Ecological Determinants of Enthusiasm in 13,661 Youth: An Integrative, Cross-sectional Analysis Using Machine Learning. *JMIR Public Health and Surveillance*.
- Tripathi, Sandhya; Fritz, Bradley A; **Abdelhack, Mohamed**; Avidan, Michael S; Chen, Yixin; and King, Christopher R. 2024. Multi-view representation learning to schema match databases. *Journal of Biomedical Informatics*
- Hall, Siobhan Mackenzie; Kochin, Daniel; Carne, Carmel; Herterich, Patricia; Lewers, Kristen Lenay; **Abdelhack, Mohamed**; Ramasubramanian, Arun; Michael Alphonse, Juno Felecia; Ung, Visotheary; El-Gebali, Sara; Currin, Christopher Brian; Plomp, Esther; Thompson, Rachel; Sharan, Malvika. 2024. Ten Simple Rules for pushing boundaries of inclusion at academic events. *PLOS Computational Biology*
- Abdelhack, Mohamed**; Zhukovsky, Peter; Milic, Milos; Harita, Shreyas; Wainberg, Michael; Tripathy, Shreejoy; Griffiths, John; Hill, Sean and Felsky, Daniel. 2023. Opposing brain signatures of sleep in task-based and resting-state conditions. *Nature Communications*
- Abdelhack, Mohamed**; Zhang, Jiaming; Tripathi, Sandhya; Fritz, Bradley A; Felsky, Daniel; Avidan, Michael S; Chen, Yixin; and King, Christopher R. 2023. A modulation layer to increase neural network robustness against data quality issues. *Transactions on Machine Learning Research*
- Cheng Fan L.*; Horikawa, Tomoyasu; Majima, Kei; **Abdelhack, Mohamed**; Aoki, Shuntaro C.; Hirano, Jin and Kamitani, Yukiyasu. 2023. Reconstructing visual illusory experience from human brain activity. *Science Advances*
- King, Christopher R et al. (**Abdelhack, Mohamed** as a part of the ACTFAST Study Group). 2023. An intraoperative telemedicine program to improve perioperative quality measures: The ACTFAST-3 randomized clinical trial. *JAMA Network Open*
- Abdelhack, Mohamed**. 2022. An in vivo assay to study locomotion in *Caenorhabditis elegans*. *MethodsX*
- Ortiz, Abigail; Sanches, Marcos; **Abdelhack, Mohamed**; Schwaiger, Tyler R; Wainberg, Michael; Tripathy, Shreejoy J; Felsky, Daniel; Mulsant, Benoit H; Fiedorowicz, Jess G. 2022. Sex-specific associations between lifetime diagnosis of bipolar disorder and cardiovascular disease: a cross-sectional analysis of 257,673 participants from the UK Biobank. *Journal of Affective Disorders*
- Hueniken, Katrina; Somé, Nibene Habib; **Abdelhack, Mohamed**; Taylor, Graham; Elton Marshall, Tara; Wickens, Christine M; Hamilton, Hayley A; Wells, Samantha; Felsky, Daniel. 2021. Machine learning-based predictive modeling of anxiety and depressive symptoms during 8 months of the COVID-19 global pandemic: repeated cross-sectional survey study. *JMIR mental health*
- Fritz, Bradley A; **Abdelhack, Mohamed**; King, Christopher R; Chen, Yixin; and Avidan, Michael S. 2020. Update to ‘Deep-learning model for predicting 30-day postoperative mortality (Br J Anaesth 2019; 123: 688–95). *British Journal of Anaesthesia*
- Abdelhack, Mohamed** and Kamitani, Yukiyasu. 2018. Sharpening of hierarchical visual feature representations of blurred images. *eNeuro*
- Nedelescu, Hermina; **Abdelhack, Mohamed**; and Pritchard, Arwel T. 2018. Regional differences in Purkinje cell morphology in the cerebellar vermis of male mice. *Journal of neuroscience research*
- Nedelescu, Hermina and **Abdelhack, Mohamed**. 2013. Comparative morphology of dendritic arbors in populations of Purkinje cells in mouse sulcus and apex. *Neural Plasticity*

UNDER REVIEW

Hara, Shuhei; Suzuki, Keita; Murakami, Akio; Majima, Kei; **Abdelhack, Mohamed**; Fan, Cheng; Doya, Kenji and Takahashi, Hidehiko. Dysregulated Loops of Visual Cortical Hierarchies in Schizophrenia: A Circular Inference Model Approach

Clifford, Kevan P; Dos Santos, Fernanda C; **Abdelhack, Mohamed**; Felsky, Daniel; Sibille, Etienne and Nikolova, Yuliya S. Neurostructural and cognitive signatures of novel polygenic risk scores for molecular brain aging

Akbarian, Niki; Ebrahimi, Mahbod; Dos Santos, Fernanda C; **Abdelhack, Mohamed**; Sanches, Marcos; Diaconescu, Andreea O; Rajji, Tarek K; Felsky, Daniel; Zai, Clement and Kennedy, James L. Depression, Hypertension, and Alcohol Consumption Mediate the Relationship between Neuroticism and Alzheimer's Disease: A UK Biobank Study

IN PREP

Abdelhack, Mohamed; Koulahdouzi, Mojtaba; Nashnoush, Eptehal; Ghoshal, Arun; Guo, Eddie; Lance, Jake; Calarco, Navona and Seyyed-Kalantari, Laleh. Fairness Challenges in Multimodal Foundational Models in Healthcare

Suzuki, Keita*; **Abdelhack, Mohamed***; Murakami, Akio; Majima, Kei; Cheng Fan; Kamitani, Yukiyasu and Takahashi, Hidehiko. Schizophrenia patients show a diffuse neural activation pattern to object recognition

Abdelhack, Mohamed*; Murakami, Akio*; Majima, Kei; Cheng Fan; Kamitani, Yukiyasu and Takahashi, Hidehiko. Analysis of top-down and bottom-up processes of visual recognition in schizophrenia using deep neural network models

Abdelhack, Mohamed; Gonzalez-Torres, Christina; Ortiz, Abigail; Goldstein, Benjamin; and Felsky, Daniel. Population-level integrative modelling of factors associated with elevated cardiovascular disease risk in women with bipolar disorder

Salah, Alaa⁺; Mostafa, Abdalrhman⁺; Abedi, Reem⁺; El Masri, Ghinwa⁺; Hammad, Ghada⁺; Eltaras, Menna⁺; Fadl, Amira⁺; Khaira, Enas⁺; Elsubaie, Rawan; **Abdelhack, Mohamed**. Arabs in Neuroscience: a case for localization in diversity educational programs

Knowledge Mobilization

Abdelhack, Mohamed. Breaking the Wall of Exclusivity in Science. *Idea Pitch at Falling Walls Lab Toronto 2024, August*.

Hill, Sean; Kloiber, Stefan; Ortiz, Abigail; Le Foll, Bernard; Felsky, Daniel; Maslej, Marta; Tripathy, Shreejoy; Tang, Victor; Yu, Joanna; **Abdelhack, Mohamed** and Sloan, Matthew. Predicting Patient Outcomes in the CAMH Major Depressive Disorder (MDD), Bipolar Disorder (BD), Alcohol Use Disorder (AUD), and AUD-MDD Integrated Care Pathways (ICPs). *Retrospective Study Protocol, Approved by Research Ethics Board: 2023, April*.

Abdelhack, Mohamed; Tripathi, Sandhya; Fritz, Bradley A; Avidan, Michael S; Chen, Yixin; and King, Christopher R. Machine learning-based decision support system to predict post-surgical mortality and acute kidney injury. *Deployed tool at Barnes Jewish Hospital, Washington University School of Medicine*. A part of the TECTONICS (Telemedicine Control Tower for the OR: Navigating Information, Care and Safety) trial. Deployed: 2021, February.

Awards, Fellowships, Honors, & Grants

2025	Rising Stars of Basic Neuroscience (TBA: November-2025) , The Transmitter	
2025	Best Presentation by a Fellow Award , Department of Psychiatry, University of Toronto	\$CA 500
2025	French Neuroscience Society – IBRO Travel Award , International Brain Research Organization	\$CA 2,800 (eq.)
2025	SOBP Travel Award , Society of Biological Psychiatry	\$CA 2,709 (eq.)
2024	Justice, Equity, Diversity and Inclusion (JEDI) Award , Life Science Editors Foundation	
2024	Biomedical Sciences Award - Declined , BioScience Scientists Awards	
2023	IBRO School Support - For Arabs in Neuroscience Initiative , International Brain Research Organization	\$CA 28,060 (eq.)
2024	Open Science Prizes - International Trainee Prize , The Neuro – Irv and Helga Cooper Foundation	\$CA 10,000
2024	Canadian Neuroanalytics Scholars Program , Campus Alberta Neuroscience	\$CA 140,000

2024	Project Grant - Fall 2023 and Spring 2024 - as co-PI , Canadian Institutes of Health Research	\$CA 1,013,626
2024	IBRO Diversity Grant - For Arabs in Neuroscience Initiative , International Brain Research Organization	\$CA 3,747 (eq.)
2024	Discovery Fund 2024 Postdoctoral Research Award Competition , Centre of Addiction and Mental Health	\$CA 65,000
2023	Brain Star Award , Canadian Association for Neuroscience and the Canadian Institutes of Health Research Institute of Neuroscience, Mental Health, and Addiction	\$CA 1,500
2023	IBRO Meetings Support - For Arabs in Neuroscience Initiative , International Brain Research Organization	\$CA 3,570 (eq.)
2023	Simons Collaboration on the Global Brain (SCGB) Conference Award Program - For Arabs in Neuroscience Initiative , Simons Foundation	\$CA 26,338 (eq.)
2022	Top Reviewer Award , Conference on Neural Information Processing Systems	
2022	womenmind™2022 Postdoctoral Fellowship Competition , Centre of Addiction and Mental Health	\$CA 65,000
2022	IBRO Diversity Grant - For Arabs in Neuroscience Initiative , International Brain Research Organization	\$CA 3,226 (eq.)
2022	Highlighted Reviewer of ICLR 2022 , International Conference on Learning Representations	
2019	Fourth Place in Image Recognition Hackathon , AI Hack Tunisia	Internship Offer
2018	Monbukagakusho Scholarship , Japanese Ministry of Education, Culture, Sports, Science and Technology	\$CA 10,335 (eq.)
2017	Monbukagakusho Scholarship , Japanese Ministry of Education, Culture, Sports, Science and Technology	\$CA 21,610 (eq.)
2017	Summer School Cost Waiver , Computational and Cognitive Neuroscience Summer School	\$CA 1,975 (eq.)
2016	Tuition Exemption Award , Kyoto University	\$CA 5,034 (eq.)
2015	Tuition Exemption Award , Kyoto University	\$CA 2,517 (eq.)
2015	Admission Fee Waiver Award , Kyoto University	\$CA 2,650 (eq.)
2010	Cairo International Biomedical Engineering Conference Registration Fee Grant , Misr El Kheir Foundation	\$CA 20 (eq.)

Presentations

* *presenting author*; + *mentored student*

INVITED TALKS

November 2025. Ramathibodi Global Collaboration Unit (RGCU), Bangkok, Thailand.

October 2025. Brain and Learning Lab, University of Geneva, Geneva, Switzerland.

September 2025. Invited panel: KCNI Open House, Toronto, Canada.

May 2025. Vlaams Instituut voor Biotechnologie (VIB), Leuven, Belgium.

May 2025. FRCB-IDIBAPS, Barcelona, Spain.

May 2025. Pompeu Fabra University, Barcelona, Spain.

Mar 2025. Department of Psychology, University of British Columbia, Canada.

Jan 2025. Center for Brain and Health (CBH) Winter Symposium. New York University Abu Dhabi, UAE.

Jan 2025. Biological and Environmental Science and Engineering Division, King Abdullah University of Science and Technology (KAUST), Saudi Arabia.

Dec 2024. Misr University for Science and Technology, Egypt.

Dec 2024. School of Biotechnology, Nile University, Egypt.

Nov 2024. Imaging a Better Tomorrow Series 2, Celebrating the International Day of Radiology Under the theme of “Illuminating the invisible”. Faculty of Applied Health Sciences Technology, Galala University, Egypt (Online).

Nov 2024. Open Science in Action Symposium. Montreal, Canada.

Nov 2024. Scientific Arab Forum Life Sciences Conference. Online.

Jan 2024. CERVO Research Seminar. Quebec, Canada.

Oct 2023. NeuroPedia Neural Engineering Webinar Series.

Aug 2023. American University in Cairo, Cairo, Egypt.

Dec 2022. Invited panel: Montreal AI & Neuroscience Conference 2022, Montreal, Canada.

Jul 2022. Simply Neurocon Conference 2022, Online.

May 2022. IEEE Engineering in Medicine and Biology Society American University of Sharjah Student Branch, UAE (Online).

Apr 2022. Egypt Scholars, Online.

Oct 2021. Invited webinar: Birzeit University, Palestine (Online).

Dec 2020. Invited webinar: Society of Neuroscientists of Africa.

PLENARY PRESENTATIONS

Abdelhack, Mohamed; Tio, Earvin S; Gonzalez-Torres, Christina; Ortiz, Abigail; and Felsky, Daniel. 2025. Revealing the Role of Medications in the Elevated Risk of Cardiovascular Disorders in Women with Bipolar Disorder Using Machine Learning. Rising Star Showcase presentation, Society of Biological Psychiatry Annual Meeting, Toronto, Canada.

Abdelhack, Mohamed*; Zhukovsky, Peter; Milic, Milos; Harita, Shreyas; Wainberg, Michael; Tripathy, Shreejoy; Griffiths, John; Hill, Sean and Felsky, Daniel*. 2024. Opposing brain signatures of sleep in task-based and resting-state conditions. University of Toronto, Department of Psychiatry Research Day, Toronto, Canada.

ORAL PRESENTATIONS

Abdelhack, Mohamed; Tio, Earvin S; Gonzalez-Torres, Christina; Ortiz, Abigail; and Felsky, Daniel. 2025. Revealing the Role of Medications in the Elevated Risk of Cardiovascular Disorders in Women with Bipolar Disorder Using Machine Learning. University of Toronto, Department of Psychiatry Research Day, Toronto, Canada.

Abdelhack, Mohamed*. 2025. Uncovering shared neural signatures of cognition in neurodegenerative disorders with neuroimaging data analytics and computational modelling. Canadian Neuroanalytics Scholars Program Meeting, Montreal, Canada.

Abdelhack, Mohamed*. 2024. Unraveling opposing neural signatures of sleep, depression, and cognition measures. Canadian Neuroscience Seminars - Postdoctoral Series (CNS-PDS).

Abdelhack, Mohamed*; Zhukovsky, Peter; Milic, Milos; Harita, Shreyas; Wainberg, Michael; Tripathy, Shreejoy; Griffiths, John; Hill, Sean and Felsky, Daniel. 2023. Sleep and depression show contradicting neural signatures during task-engaged and resting states. Canadian Computational Neuroscience Spotlight, Online.

Abdelhack, Mohamed*; Zhang, Jiaming; Tripathi, Sandhya; Fritz, Bradley A; Felsky, Daniel; Avidan, Michael S; Chen, Yixin; and King, Christopher R. 2023. Bio-inspired modulation to enhance the robustness of deep learning models against healthcare data quality problems. T-CAIREM AI in Medicine Conference. Toronto, Canada.

Abdelhack, Mohamed*; Zhukovsky, Peter; Milic, Milos; Harita, Shreyas; Griffiths, John; Hill, Sean and Felsky, Daniel. 2023. Sleep and depression show contradicting neural signatures during task-engaged and resting states. University of Toronto, Department of Psychiatry Research Day. Toronto, Canada.

Salah, Alaa⁺; Mostafa, Abdalrhman⁺; Abedi, Reem⁺; El Masri, Ghinwa⁺⁺; Hammad, Ghada⁺; Eltaras, Menna⁺; Fadl, Amira⁺; Khaira, Enas⁺; **Abdelhack, Mohamed**. 2022. Computational neuroscience in the arabic region. Flash talk: Neuromatch Conference 5.0.

Elsherif, Omar⁺⁺, **Abdelhack, Mohamed**. 2021. Modeling image sharpening in human visual cortex using computer vision models. Flash talk: Neuromatch Conference 4.0.

Abdelhack, Mohamed*. 2021. Big data in the wild: modulation of artificial intelligence model weights in real time. Flash talk: Krembil Centre for Neuroinformatics Open House.

Abdelhack, Mohamed*. 2021. Modulation layer for combating missing data. Anesthesiology Research Data Blitz, Washington University in St. Louis, St. Louis, MO.

Abdelhack, Mohamed* and Kamitani, Yukiyasu. 2020. Visual misrecognition reveals contradiction between lower and higher visual cortex activations. Flash talk: Neuromatch Conference 3.0.

POSTER PRESENTATIONS

Abdelhack, Mohamed*; Wickramatunga, Rajith and Felsky, Daniel. 2025. Longitudinal co-trajectories of neural dynamics underlying sleep and depression. Poster presentation: Neuronto Research Symposium, Toronto, Canada.

Abdelhack, Mohamed*; Wickramatunga, Rajith and Felsky, Daniel. 2025. Longitudinal co-trajectories of neural dynamics underlying sleep and depression. Poster presentation: Organization of Human Brain Mapping Annual Meeting, Brisbane, Australia.

Abdelhack, Mohamed*; Wickramatunga, Rajith and Felsky, Daniel. 2025. Longitudinal co-trajectories of neural dynamics underlying sleep and depression. Poster presentation: French Society for Neuroscience Annual Meeting, Montpellier, France.

Abdelhack, Mohamed*; Wickramatunga, Rajith and Felsky, Daniel. 2025. Longitudinal co-trajectories of neural dynamics underlying sleep and depression. Poster presentation: Society of Biological Psychiatry Annual Meeting, Toronto, Canada.

Verma, Siya*; Kloiber, Stefan; Dela Cruz, Gil Angela; Aloysius, Ryan; **Abdelhack, Mohamed**; Morera-Gonzalez, Daniela; Santisteban, Jose Arturo; Felsky, Daniel; Tripathy, Shreejoy; Jones, Brett; Tang, Victor; Maslej, Marta; Levitan, Robert and Hill, Sean. 2025. Indicators of Major Depressive Disorder (MDD) Subgroups Within the Major Depressive Disorder-Integrated Care Pathway (MDD-ICP). Poster presentation: Society of Biological Psychiatry Annual Meeting, Toronto, Canada.

Tio, Earvin S; **Abdelhack, Mohamed** and Felsky, Daniel. 2025. Understanding the Biopsychosocial Mechanisms of Risk for Suicide Using Machine Learning and a Resilience Framework. Poster presentation: Society of Biological Psychiatry Annual Meeting, Toronto, Canada.

Hara, Shuhei*; Suzuki, Keita; Murakami, Akio; Majima, Kei; **Abdelhack, Mohamed**; Cheng, Fan; Hidehiko, Takahashi and Doya, Kenji. 2025. fMRI Study Reveals Enhanced Top-down and Bottom-up Processes in Schizophrenia. Poster presentation: Computational and Systems Neuroscience (COSYNE) conference, Montreal, Canada.

Huang, Lu*; **Abdelhack, Mohamed** and Felsky, Daniel. 2024. Detection of Cerebrovascular Diseases with in Vivo MRI and Deep Neural Networks. Poster presentation: University of Toronto, Institute of Medical Sciences Summer Undergraduate Research Program Research Day (Best Presenter Award to Huang, Lu).

Abdelhack, Mohamed*; Wickramatunga, Rajith and Felsky, Daniel. 2024. Longitudinal co-trajectories of neural dynamics underlying sleep and depression. Poster presentation: Society for Neuroscience Annual Meeting, Chicago, IL.

Bakkar, Nour Mounira*; Mostafa, Abdalrhman*; Eladl, Ahmed K.; **Abdelhack, Mohamed**; El-Yazbi, Ahmed F. 2024. Age- and Sex- Specific Aberrations of Arterial Pressure Control in Prediabetes: What does Machine Learning reveal about Frequency Domain Analysis of Blood Pressure?. Poster presentation: ASPET 2024 Annual Meeting. Arlington, VA.

Abdelhack, Mohamed*; Wickramatunga, Rajith and Felsky, Daniel. 2024. Shared longitudinal neural representations of sleep, depression, and cognition. Poster presentation: Society for Cognitive Neuroscience Annual Meeting, Toronto, Canada.

Abdelhack, Mohamed*; Murakami, Akio; Suzuki, Keita; Majima, Kei; Cheng Fan; Kamitani, Yukiyasu and Takahashi, Hidehiko. 2023. Uncovering the Visual Processing Imbalance in Schizophrenia Using Deep Neural Network Representations. Poster presentation: The New VISTAs in Vision Research, Vision Research Conference, Toronto, Canada.

Mostafa, Abdalrhman*; Bakkar, Nour Mounira*; **Abdelhack, Mohamed**; El-Yazbi, Ahmed F. 2023. Identification of Novel Spectral Densities in Arterial Pressure Time Series for Cardiovascular Risk Stratification in Early Prediabetes: Machine Learning Beyond Conventional Frequency Domain Analysis. Poster presentation: American Heart Association Scientific Sessions 2023.

Hoxha, Isabelle; Mudrik, Noga; Urai, Anne; Kienigiel, Dante; Forest, Jeremy; **Abdelhack, Mohamed**; Peters, Megan A K; Halper, Nick; Zhang, Ru-Yuan; Lu, Xinquan; Butler, John S.* 2023. Opening Computational Neuroscience to a Wider Audience: Virtual Escape Room for Kids. Poster presentation: Conference on Cognitive Computational Neuroscience. Oxford, United Kingdom.

- Abdelhack, Mohamed***; Zhukovsky, Peter; Milic, Milos; Hill, Sean; Felsky, Daniel. 2023. Unraveling conflicting neural signatures of subjective and objective sleep measures. Poster presentation: Organization for Human Brain Mapping Annual Meeting. Montreal, Canada.
- Hara, Shuhei*; Suzuki, Keita; Murakami, Akio; Majima, Kei; **Abdelhack, Mohamed**; Cheng Fan; Takahashi, Hidehiko and Doya, Kenji. 2023. Elucidation of the dysfunction in sensory processing in schizophrenia. Poster presentation: Computational Psychiatry Conference, Dublin, Ireland.
- Bakkar, Nour Mounira*; Mostafa, Abdalrhman*; **Abdelhack, Mohamed**; El-Yazbi, Ahmed F. 2023. A convolutional neural network model classifies beat-to-beat arterial pressure time-series: sex stratification and cardiovascular risk prediction. Poster presentation: ASPET 2023 Annual Meeting. St. Louis, MO.
- Abdelhack, Mohamed***; Zhukovsky, Peter; Milic, Milos; Hill, Sean; Felsky, Daniel. 2023. Shared Neural Representations between Sleep and Depression and Their Effect on Visual Perception. Poster presentation: Society of Biological Psychiatry Annual Meeting. San Diego, CA.
- Dolling-Boreham, Roberta M*; Mohan, Akshay; **Abdelhack, Mohamed**; Elton-Marshall, Tara; Hamilton, Hayley A; Boak, A; Felsky, Daniel. 2023. Identifying psychosocial and ecological determinants of enthusiasm in youth using machine learning. Poster presentation: International Conference on Health Policy Statistics. Scottsdale, AZ.
- Hoxha, Isabelle; Mudrik, Noga; Urai, Anne; Kienigiel, Dante; Forest, Jeremy; **Abdelhack, Mohamed**; Peters, Megan A K; Zhang, Ru-Yuan; Lu, Xinquan; Halper, Nick; Butler, John S.* 2022. Virtual escape room for kids: opening online conferences to a wider audience. Poster presentation: SCI:COM
- Abdelhack, Mohamed***; Zhukovsky, Peter; Milic, Milos; Felsky, Daniel. 2022. Effect of sleep on neural coding in early visual cortex revealed in a large population study. Poster presentation: Society for Neuroscience Annual Meeting, San Diego, CA.
- Abdelhack, Mohamed***; Zhukovsky, Peter; Milic, Milos; Felsky, Daniel. 2022. Effect of sleep on neural coding in early visual cortex revealed in a large population study. Poster presentation: Krembil Centre for Neuroinformatics Open House.
- Abdelhack, Mohamed***; King, Christopher R; Tripathi, Sandhya; Fritz, Bradley; Chen, Yixin; Avidan, Michael. 2020. Post-operative mortality prediction with and without the use of intraoperative features. Poster presentation: AMIA 2020 Symposium.
- Abdelhack, Mohamed*** and Kamitani, Yukiyasu. 2019. Top-down modulation in human visual cortex revealed by brain decoding with deep neural network representation. Poster presentation: CiNet Conference, Osaka, Japan.
- Murakami, Akio*; **Abdelhack, Mohamed***; Majima, Kei; Cheng Fan; Kamitani, Yukiyasu and Takahashi, Hidehiko. 2019. Representational structure of the higher visual cortex in schizophrenia. Poster presentation: Mechanism of Brain and Mind Winter Workshop, Hokkaido, Japan.
- Murakami, Akio*; **Abdelhack, Mohamed***; Majima, Kei; Cheng Fan; Kamitani, Yukiyasu and Takahashi, Hidehiko. 2018. Analysis of top-down and bottom-up processes of visual recognition in schizophrenia using deep neural network models. Poster presentation: International Consortium on Hallucination Research, Kyoto, Japan.
- Abdelhack, Mohamed*** and Kamitani, Yukiyasu. 2017. Top-down neural processing that supplements missing image features revealed by brain decoding with deep neural network representation. Poster presentation: Vision Science Society Annual Meeting, St. Pete Beach, FL.
- Nedelescu, Hermina*; **Abdelhack, Mohamed**; and Pritchard, Arwel T. 2016. Purkinje cell heterogeneity in the anterior and posterior cerebellar lobule. Dynamic poster presentation: Society for Neuroscience Annual Meeting, San Diego, CA.
- Abdelhack, Mohamed*** and Maruyama, Ichiro. 2015. Locomolyzer: a comprehensive tool for animal locomotion analysis. Poster presentation: Congress of the European Society of Biomechanics, Prague, Czechia.
- Abdelhack, Mohamed***. 2014. Adaptation in locomotion behavior of *Caenorhabditis elegans*. Poster presentation: Okinawa Computational Neuroscience Course, Okinawa, Japan.
- Abdelhack, Mohamed*** and Nedelescu, Hermina. 2013. Investigating the effect of anatomical location on Purkinje cell morphology. Poster presentation: Society for Neuroscience Annual Meeting, San Diego, CA.

Teaching Experience

Oct 2025	Ibn Sina Neurotech Autumn School , Lecturer	Abu Dhabi, UAE
Oct 2025	AI for Biomedical Engineering Graduate Course , Guest Lecturer	University of Toronto
Sep 2025	Introduction to Computational Neuroscience , Lecturer	Online
Sep 2024	Ibn Sina Neurotech Summer School , Lecturer	Cairo, Egypt
Aug 2024	Introduction to Computational Neuroscience , Lecturer	Online
Mar 2024	Al Quds University, Faculty of Medicine, Neuroscience Course, Neurophysiology of Vision Lecture , Lecturer	Remote
Jan 2024	Computational Neuroscience Imbizo , Teaching Assistant	Cape Town, South Africa
Sep 2023	American Academy of Neurology - Egypt Student Chapter, Statistics Bootcamp , Instructor	Online
Aug 2023	Introduction to Computational Neuroscience , Lecturer	Online
Jun-Jul 2023	KCNI Summer Academy , Instructor	Toronto, Canada
Nov 2022	IBRO-ARC School on Impact of Lifestyle Modification on Neurodegenerative Disorders from Preventive and Therapeutic Perspectives , Instructor	Mansoura, Egypt
Aug 2022	IBRO-Simons Computational Neuroscience Imbizo , Teaching Assistant	Cape Town, South Africa
Jun 2022	Introduction to Computational Neuroscience , Lecturer and Course Lead	Online
Jul 2020	Neuromatch Academy , Lead Teaching Assistant	Online
2013	An Introduction to Interdisciplinary Science Workshop , Instructor & organizer	Okinawa, Japan
Jul 2009 & Jun 2012	IEEE Alexandria Student Branch , C Programming Instructor	Alexandria, Egypt

Mentoring and Supervision

2025	Ayesha Syeda , Summer Undergraduate Research Program, University of Toronto	Toronto, ON
2025	Mohamed Sayed , Collaboration, Dartmouth University	Online
2025	Lillian Dong , Summer Practicum Program, Department of Biostatistics, University of Toronto	Toronto, ON
2024	Alameen Saleem , Summer Undergraduate Research Program, Krembil Centre for Neuroinformatics	Toronto, ON
2024	Lu Huang , Summer Undergraduate Research Program, Krembil Centre for Neuroinformatics	Toronto, ON
2023	Mai Gamal , Visiting Student, Krembil Centre for Neuroinformatics	Toronto, ON
2022-2023	Cathlyn Chen , Research Student, Krembil Centre for Neuroinformatics	Toronto, ON
2021-2023	Roberta M Dolling-Boreham , Research Student, Krembil Centre for Neuroinformatics	Toronto, ON
2021	Katrina Heuniken , MSc Student, Krembil Centre for Neuroinformatics	Toronto, ON
2021-	Abdalrhman Mostafa , Mentee, Arabs in Neuroscience	Online
2021	Omar ElSharif , Mentee, Arabs in Neuroscience	Online

2020-2021	Jiaming Zhang , Intern, Department of Computer Science, Washington University in St. Louis	Online
2020-2021	Jerome Gammie , Intern, Department of Computer Science, Washington University in St. Louis	St. Louis MO
2017-2019	Shirin Vafae , Intern/MSc student, Graduate School of Informatics, Kyoto University	Kyoto, Japan
2018-2019	Fan Cheng , Intern/PhD Student, Graduate School of Informatics, Kyoto University	Kyoto, Japan
2018-2019	Ken Shirakawa , Undergraduate thesis student, Graduate School of Informatics, Kyoto University	Kyoto, Japan
2018-2019	Emi Satake , Master student, Graduate School of Informatics, Kyoto University	Kyoto, Japan

Media Appearances

2024	The Brainy Brain , Between Computer and Brain	Apple Podcast
2022	PhD - The Philosophical Drama , Cultivating Community in Academia	Instagram
2020	Washington University Postdoctoral Society , Postdocs Go Live	St. Louis, MO
2018	Sciency Daily , Neuro filter sharpens visual processing	Article
2016	TEDx Kyoto University , Conscious food choices for everyone	Kyoto, Japan

Outreach & Professional Development

SERVICE AND OUTREACH

2024	CAMH Postdocs , Founder	Toronto, Canada
2025	Neuro – Irv and Helga Cooper Foundation Open Science Prizes , Prize Selection Committee Member	Montreal, Canada
2024-2025	Canadian Neuroscience Seminars – Postdoctoral Series , Committee Member	Canada
2024	Arabs in Neuroscience Social at Society for Neuroscience Conference , Chair	Chicago, United States
2023	CIHR - Bias in Peer Review Training Module , Reviewer	Canada
2022	Neuromatch Conference 5.0 , Organizer	Online
2020-	Arabs in Neuroscience , Founder	Canada
2021-	The Africa I Know , Mentorship Team Member	Online
2021	Neuromatch Conference 4.0 , Organizer	Online
2021	Neuromatch Academy , Chief Communication Officer & Outreach Coordinator	Online
2021	Neuromatch Academy Computational Neuroscience Summer School , Project Mentor	Online
2019-2021	Washington University Postdoc Society , Policy Committee Member	St. Louis MO
2011	Brain-Computer Interfacing Challenge , Judging Panel Member	Albany, NY

DEVELOPMENT

Careers in Neuroscience Panel Discussion, invited by Alexandria Neuroscience Committee.

Computational and Cognitive Neuroscience Summer School 2017, New York University Shanghai, China.

OIST Computational Neuroscience Course 2014, Okinawa, Japan.

PEER REVIEW

Journals

Nature Human Behavior

Nature Computational Science

Biological Psychiatry: Cognitive Neuroscience and Neuroimaging

Imaging Neuroscience

Transactions on Machine Learning Research

PLOS One

British Journal of Anaesthesia

American Journal of Infection Control

Journal of Biomedical Informatics

Brain Imaging and Behavior

Conferences

The Conference on Neural Information Processing Systems (NeurIPS)

International Conference on Learning Representations (ICLR)

The International Joint Conference on Artificial Intelligence (IJCAI)

IEEE Big Data

PROFESSIONAL MEMBERSHIPS

Temerty Centre for AI Research and Education in Medicine

Organization for Human Brain Mapping

Society for Neuroscience

Cognitive Neuroscience Society